



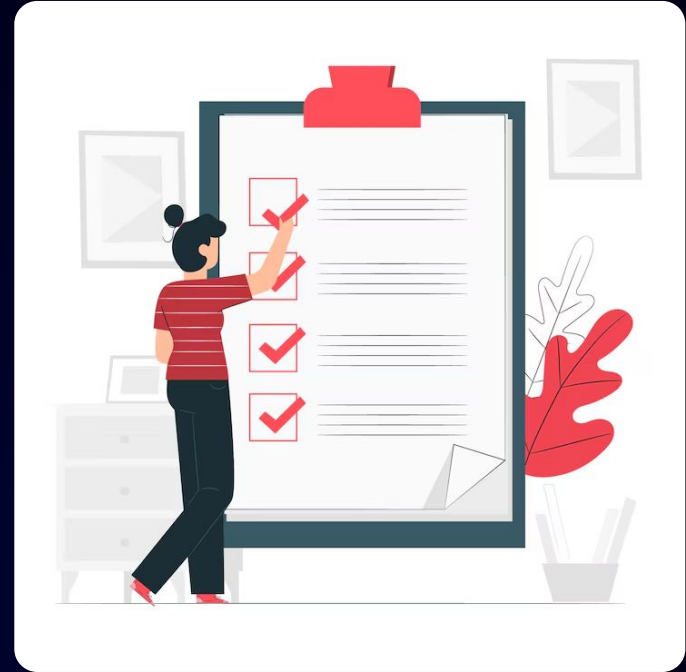
CSS Basics – Styling Your Web Pages

Learning Objectives

- Understand the purpose of CSS and how it is used to style HTML elements.
- Learn how to apply basic CSS using selectors (element, class, and id).
- Use colors and background properties to enhance the visual appearance of a webpage.
- Apply typography styles such as fonts, sizes, and text alignment to improve readability.
- Implement CSS styles in an existing HTML project through hands-on practice.

Prerequisites

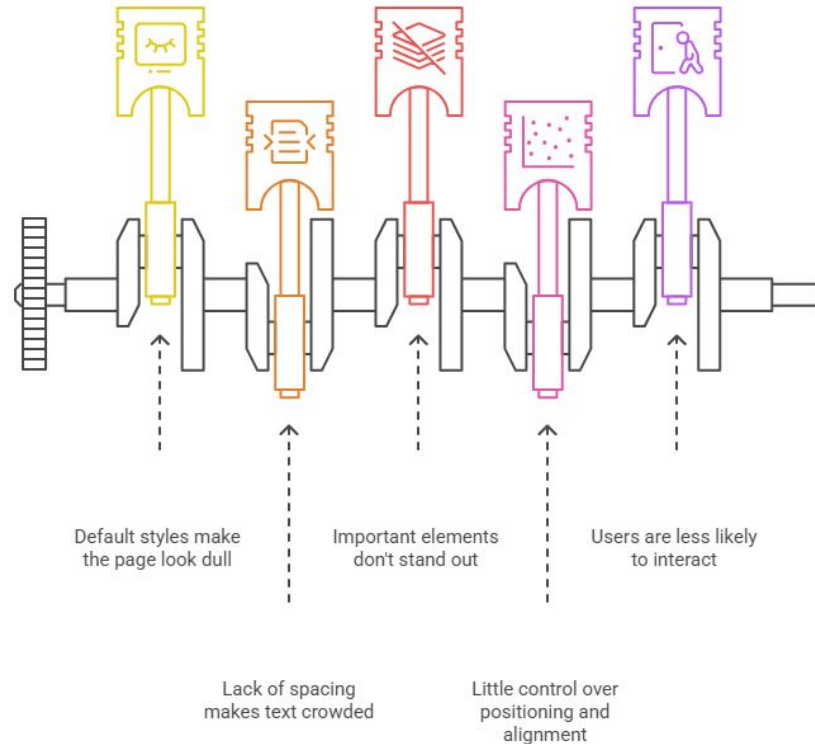
- Basic understanding of programming
- Basic understanding of HTML





Why CSS is Needed

Problem with Plain HTML (No Styling)



CSS ?

What is CSS?

- CSS stands for Cascading Style Sheets and is a powerful styling language used to control the presentation and layout of web content. It works alongside HTML to define how elements should appear on a webpage. With CSS, you can set colors, fonts, spacing, borders, and more.
- It's a language specifically designed to style the presentation of a web page.
- Offers greater control over layout, fonts, colors, and more.

```
/* Sample CSS */
body {
  font-family: Arial, sans-serif;
  background-color: #f9f9f9;
}

h1 {
  color: #333;
  text-decoration: underline;
}

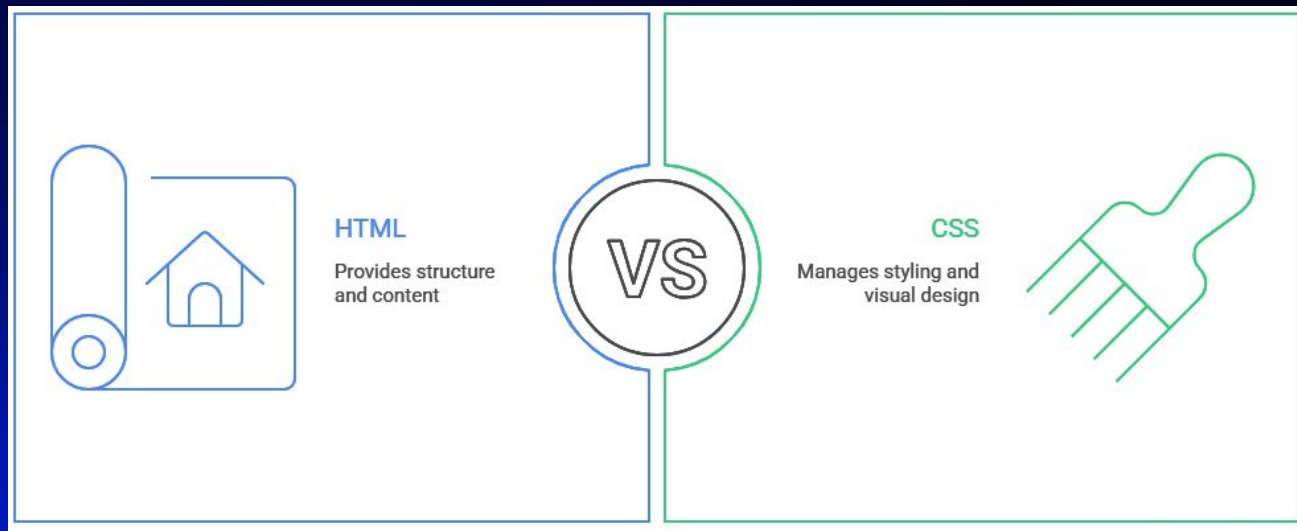
p {
  font-size: 16px;
  line-height: 1.5;
}
```

Separation of concerns

In simple term, HTML builds the house, CSS decorates it.

Keeping them separate makes your code:

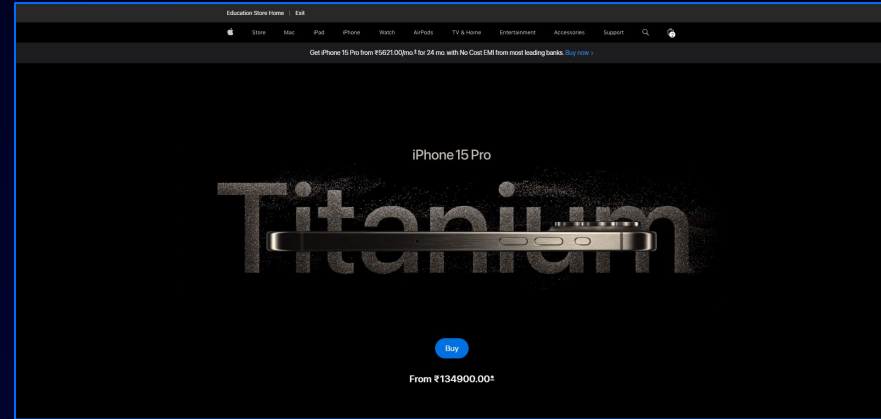
- Easier to read and maintain
- Reusable across multiple pages
- Flexible to update design without changing structure



Introduction to CSS



Without CSS



With CSS

→ *CSS improves user experience by making web pages visually appealing, easy to read, and more engaging for users.*

Pop Quiz

Q. What is the main problem with plain HTML?

A

It lacks styling and is hard to read

B

It has no structure

Pop Quiz

Q. What is the main problem with plain HTML?

A

It lacks styling and is hard to read

B

It has no structure



Ways to Add CSS

3 Approaches –

Internal Stylesheets

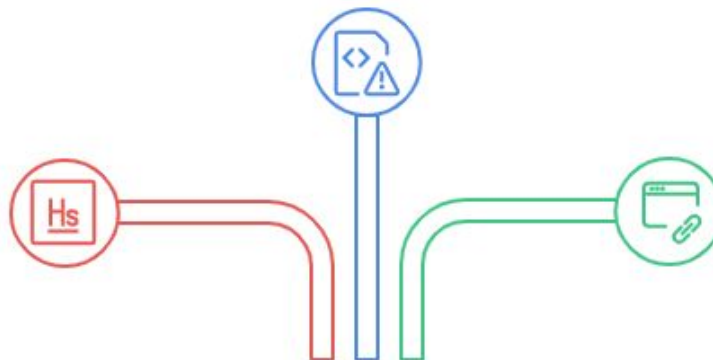
Useful for applying styles to a single HTML document, but not reusable across multiple pages.

Inline Styles

Best for applying unique styles to specific elements, but can make HTML cluttered.

External Stylesheets

Ideal for applying styles to multiple HTML documents, promoting reusability and organization.



Inline CSS

Inline CSS involves applying styles directly within an HTML element using the style attribute.

```
<!DOCTYPE html>
<html>
  <body>
    <h1 style="color:blue;">A Blue Heading</h1>
    <p style="color:red;">A red paragraph.</p>
  </body>
</html>
```

A Blue Heading

A red paragraph.

Internal CSS

An internal CSS is defined in the `<head>` section of an HTML page, within a `<style>` element.

```
<!DOCTYPE html>
<html>
<head>
<style>
body {background-color: powderblue;}
h1   {color: blue;}
p    {color: red;}
</style>
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

This is a heading

This is a paragraph.

External CSS

An internal CSS is defined in the `<head>` section of an HTML page, within a `<style>` element.

index.html

```
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="styles.css">
</head>
<body>

  <h1>This is a heading</h1>
  <p>This is a paragraph.</p>

</body>
</html>
```

+

styles.css

```
body {background-color: powderblue;}
h1   {color: blue;}
p    {color: red;}
```

=

This is a heading

This is a paragraph.

→ *External CSS is preferred in real projects because it keeps styling separate from HTML, making the code more organized, reusable, and easier to maintain.*

Hands-On Demo | Ways to Add CSS

Pop Quiz

Q. How do you link an external CSS file to HTML?

A

```
<css link="style.css">
```

B

```
<link rel="stylesheet" href="style.css">
```

Pop Quiz

Q. How do you link an external CSS file to HTML?

A

```
<css link="style.css">
```

B

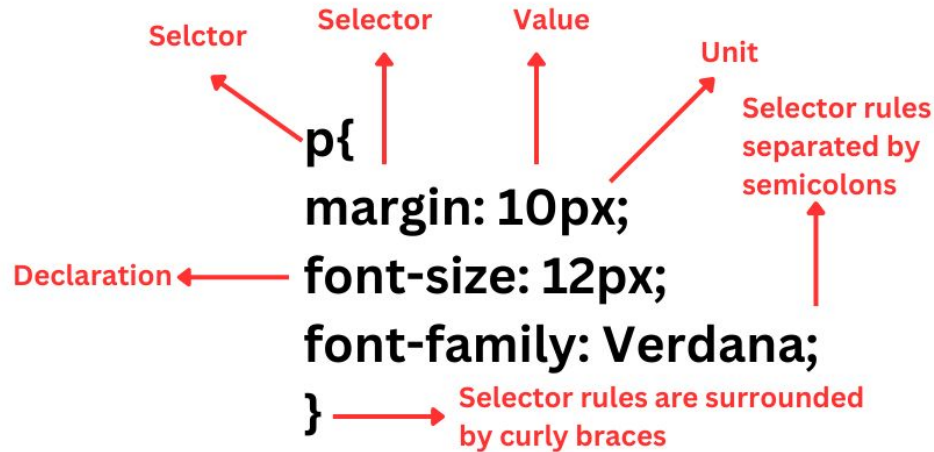
```
<link rel="stylesheet" href="style.css">
```



CSS Selectors (Core Skill)

Structure of CSS ruleset

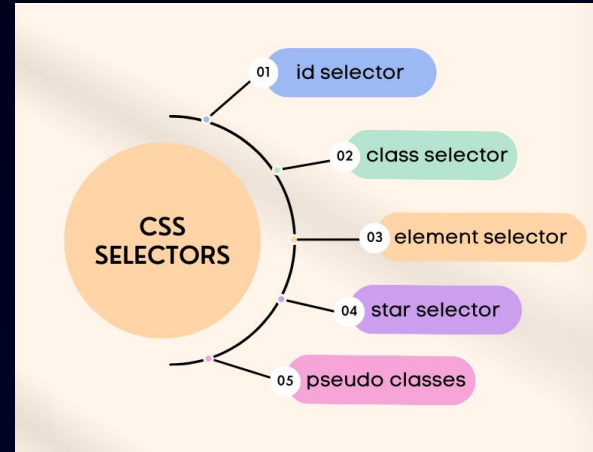
Before learning the different types of selectors in CSS, let's understand the basic structure of CSS.



Introduction to CSS Selectors

Importance of CSS Selectors

- CSS selectors are fundamental for targeting HTML elements and applying styles.
- They allow developers to specify which elements in an HTML document should receive specific styling.



Introduction to CSS Selectors

Types of CSS Selectors

- **Tagname Selectors:** Target specific HTML tags like `<p>`, `<h1>`, etc.
- **ID Selectors:** Apply unique styles to individual elements identified by a unique ID attribute.
- **Class Selectors:** Style multiple elements that share the same class attribute.
- **Universal Selectors (*):** Apply styles to all elements on a page.
- **Grouping Selectors:** Combine multiple selectors to apply the same styles to different elements.

Tagname Selectors

Using Tagname Selectors

- Tagname selectors style all elements of a specific HTML tag type uniformly.
- Useful for applying global styles to common elements like paragraphs, headings, etc.

```
/* Example: Styling all paragraphs */  
p {  
  font-size: 16px;  
  color: #333;  
  line-height: 1.5;  
}
```

ID Selectors

Applying ID Selectors

- IDs should be unique per HTML document.
- Allows precise targeting and styling of individual elements.

```
/* Example: Styling a specific element by its ID */
#header {
  background-color: #f0f0f0;
  padding: 10px;
  text-align: center;
}
```


Class Selectors

Using Class Selectors

- Classes can be applied to multiple elements, allowing for consistent styling across different elements.

```
/* Example: Styling elements with a common class */  
.button {  
  background-color: #007bff;  
  color: #fff;  
  padding: 10px 20px;  
  border-radius: 5px;  
  text-decoration: none;  
}
```

Hands-On Demo | CSS Selectors (Core Skill)

Pop Quiz

Q. How do you apply an ID in HTML?

A

```
<p id="unique">
```

B

```
<p class="unique">
```

Pop Quiz

Q. How do you apply an ID in HTML?

A

```
<p id="unique">
```

B

```
<p class="unique">
```



Grouping and Universal Selectors

Universal Selectors

Applying Universal Selectors

- “*” targets all elements in the document.
- Useful for resetting default styles or applying global styles.

```
/* Example: Applying a universal margin reset */  
* {  
  margin: 0;  
  padding: 0;  
  box-sizing: border-box;  
}
```

Grouping Selectors

- Grouping selectors minimize redundancy and improve maintainability.
- Apply shared styles to multiple elements efficiently.

```
/* Example: Grouping selectors to apply common styles */  
h1, h2, h3 {  
  font-family: 'Arial', sans-serif;  
  color: #333;  
}
```

Hands-On Demo | Grouping and Universal Selectors

Q. Which is the correct grouping selector syntax?

A

`h1.p { }`

B

`h1, p { }`

Q. Which is the correct grouping selector syntax?

A

`h1.p { }`

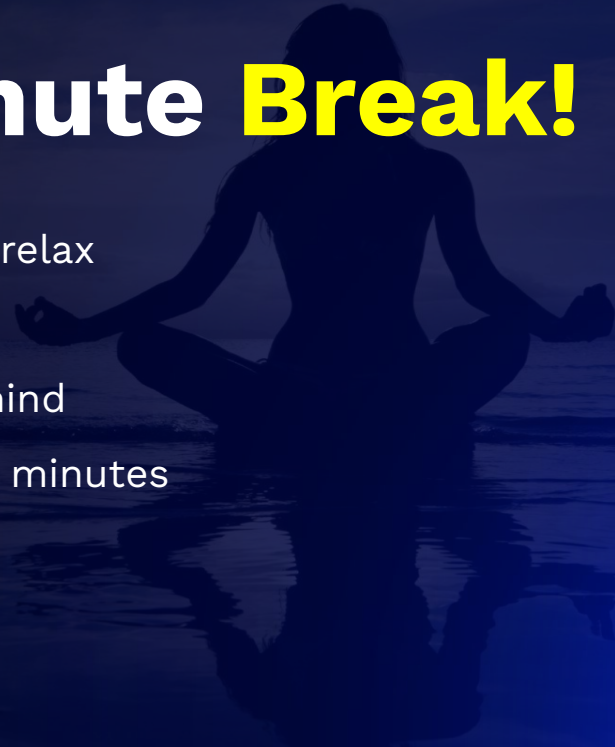
B

`h1, p { }`



Take A 5-Minute **Break!**

- Stretch and relax
- Hydrate
- Clear your mind
- Be back in 5 minutes



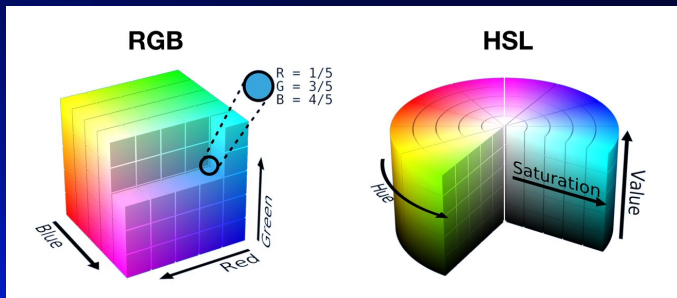


CSS Color Models

CSS Color Models

Color Models Overview

- **Color Names:** Basic predefined colors (red, blue, green).
- **Hexadecimal (Hex):** #RRGGBB format (#007bff).
- **RGB:** Red, Green, Blue values (rgb(0, 123, 255)).
- **RGBA:** RGB with an alpha channel for transparency (rgba(0, 123, 255, 0.5)).
- **HSL:** Hue, Saturation, Lightness (hsl(210, 90%, 50%)).
- **HSLA:** HSL with alpha channel (hsla(210, 90%, 50%, 0.5)).



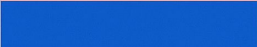
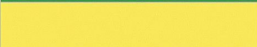
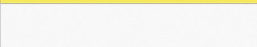
Named Colors

→ Use predefined color names

```
p {  
  color: red;  
}
```

HTML Colors

Color names supported by all browsers

Color Name	Color
Red	
Blue	
Green	
Yellow	
Black	
White	
Gray	
Cyan	
Magenta	
Orange	
Purple	
Lime	
Silver	

Hex Colors (#)

→ Use hexadecimal values

```
p {  
  color: #ff0000;  
}
```

HEXADECIMAL COLOR CODES

Red
Color
Code

Green
Color
Code

Blue
Color
Code

RR GG BB

Signifies Hex Color Code

Each of the two-digit color codes
are in the range 00...FF hexadecimal

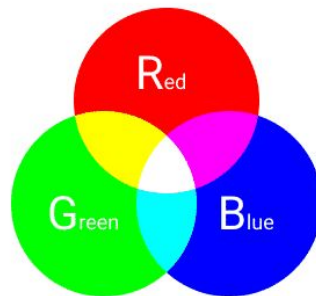
EXAMPLES

#FFCC99 #3C00A6

RGB Colors

→ Use Red, Green, Blue values

```
p {  
  color: rgb(255, 0, 0);  
}
```



HEX code

FF **FF** **FF**
Red Green Blue
00 ~ FF 00 ~ FF 00 ~ FF

RGB

rgb(**255**, **255**, **255**)
Red Green Blue
0~255 0~255 0~255

RGBA

rgba(**255**, **255**, **255**, **1.0**)
Alpha
0.0~1.0

Styling with Colors

Applying Color Styles

- Use color, background-color, and opacity properties to enhance design and readability.

```
/* Example: Setting text and background colors */  
body {  
  color: #333;  
  background-color: #f0f0f0;  
}
```

Hands-On Demo | CSS Color Models

Pop Quiz

Q. Which of the following is a named color in CSS?

A

`rgb(255, 0, 0)`

B

`red`

Pop Quiz

Q. Which of the following is a named color in CSS?

A

`rgb(255, 0, 0)`

B

`red`



Background Styling

Background Techniques

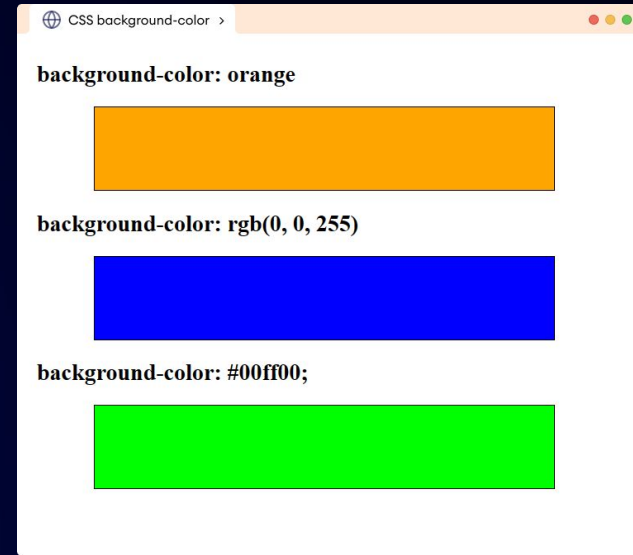
- Apply backgrounds using images, gradients, colors, and patterns.
- Use background-image, background-size, background-position, and background-repeat properties.

```
/* Example: Using a background image */  
.header {  
    background-image: url('header-bg.jpg');  
    background-size: cover;  
    background-position: center;  
}
```

Background Color

- Used to set the background color of an element

```
body {  
  Background-color: orange;  
}
```



Background Image

→ Used to set an image as background

```
body {  
  background-image: url("image.jpg");  
}
```


Hands-On Demo | Background Styling

Pop Quiz

Q. Which of the following is a named color in CSS?

A

`rgb(255, 0, 0)`

B

`red`

Pop Quiz

Q. Which of the following is a named color in CSS?

A

`rgb(255, 0, 0)`

B

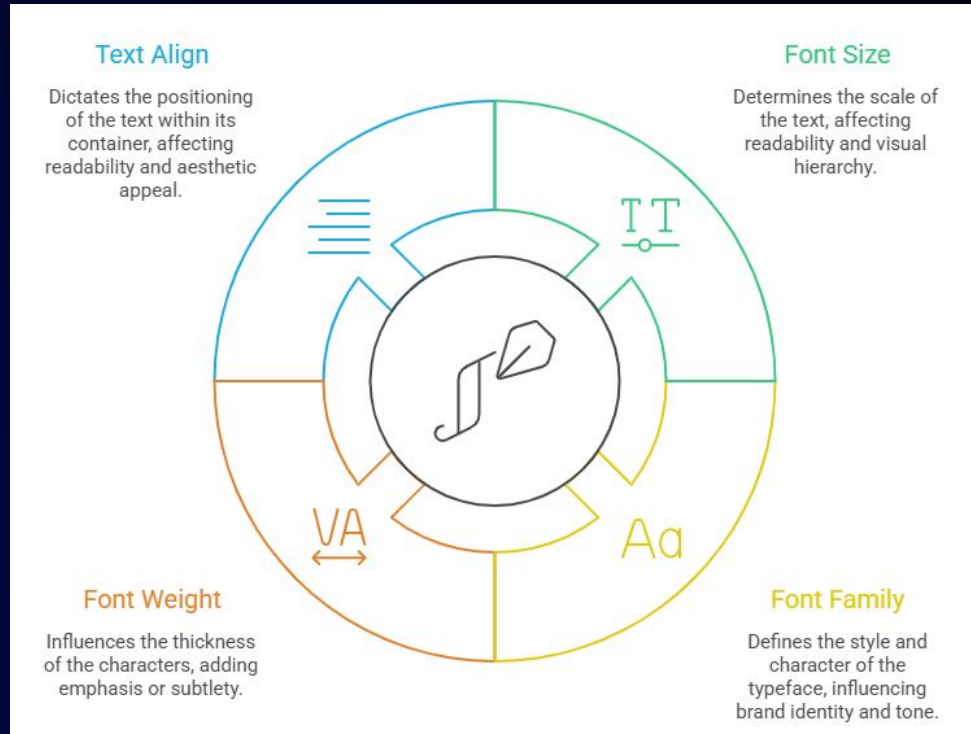
`red`



Typography Basics

Typography Overview

- Typography in CSS is the styling of text to make it readable and visually appealing.



font-size

- Controls the size of the text.

```
p {  
font-size: 18px;  
}
```

A vertical list of text samples demonstrating different font sizes. Each line is preceded by a label 'font-size: [value]' in a small font. The values range from 20px at the top to 13px at the bottom. The text size decreases as the value decreases, providing a visual scale.

font-size: 20px
font-size: 3px
font-size: 4px
font-size: 5px
font-size: 6px
font-size: 7px
font-size: 8px
font-size: 9px
font-size: 10px
font-size: 11px
font-size: 12px
font-size: 13px

font-family

- Changes the type/style of the font.

```
p {  
  font-family: Arial, sans-serif;  
}
```



font-weight

- Controls how bold or light the text appears.

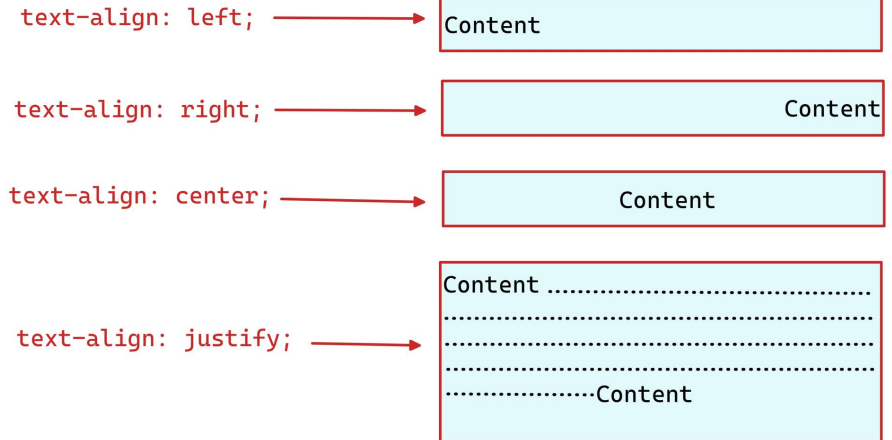
```
h1 {  
  font-weight: bold;  
}
```

font-weight: 100;	abc	}	Arial Regular
font-weight: 200;	abc		
font-weight: 300;	abc		
font-weight: 400;	abc		
font-weight: 500;	abc		
font-weight: 600;	abc	}	Arial Bold
font-weight: 700;	abc		
font-weight: 800;	abc	}	Arial Black
font-weight: 900;	abc		

text-align

- Aligns the text horizontally.

```
h1 {  
  text-align: center;  
}
```



Hands-On Demo | Typography Basics

Pop Quiz

Q. What does this code do?

```
h1 {  
text-align: center;  
}
```

A

Changes color

B

Centers the text

Pop Quiz

Q. What does this code do?

```
h1 {  
text-align: center;  
}
```

A

Changes color

B

Centers the text

Class Project

Apply Styling to Project

Summary

In this lecture, we have covered:

- Why CSS is Needed
- Different ways to CSS
- CSS Selectors
- Grouping and Universal Selectors
- Colors in CSS
- Background Properties and
- Basic of Typography in CSS.

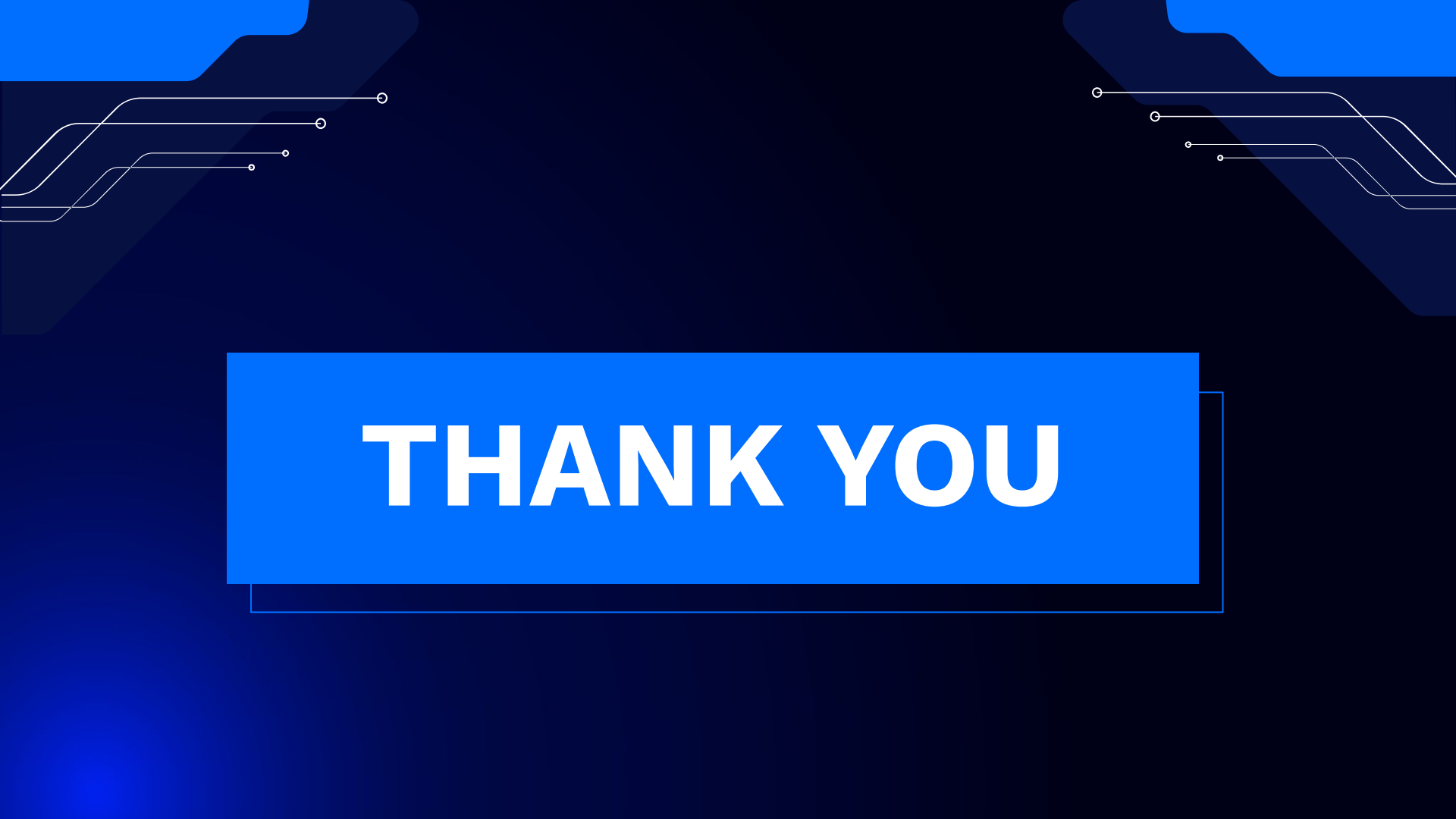




DOUBT RESOLUTION

Important

- Complete the post-class assessment
- Complete assignments (if any)
- Practice the concepts and techniques taught in this session
- Review your lecture notes
- Note down questions and queries regarding this session and consult the teaching assistants.



THANK YOU